

1. Problem Statement

Hospital readmissions within 30 days remain one of the most persistent inefficiencies in healthcare systems, particularly among diabetic patients where disease management is inherently complex and often unstable. While hospitals collect extensive patient data, the challenge is not availability but the ability to transform that data into actionable insight before discharge.

This project was therefore designed around a focused and clinically grounded objective: to identify patients at high risk of 30-day readmission while minimizing missed cases. This reframing is important because the cost of error is not symmetric. Missing a high-risk patient carries significant clinical and financial consequences, whereas incorrectly flagging a low-risk patient results in additional but manageable monitoring.

As a result, the modelling approach was intentionally built around **recall as the primary metric**, ensuring that the system prioritizes sensitivity to risk over strict classification accuracy.

2. Data Overview and Wrangling

The dataset initially contained **101,766 hospital encounters across 50 variables**, providing a comprehensive view of patient demographics, clinical conditions, and hospital interactions. However, transforming this raw dataset into a usable analytical structure required careful handling of missing values, inconsistent categories, and non-informative identifiers.

Placeholder values such as "?" were systematically converted into missing values to prevent distortion in categorical analysis. Identifier variables were removed from modelling to avoid introducing artificial relationships. The target variable, originally expressed across three categories (NO, >30, <30), was redefined into a binary outcome focused specifically on early readmission.

Following these transformations and extensive feature engineering, the dataset evolved into **99,343 records with 114 structured features**, reflecting a shift from raw inputs to meaningful clinical signals.

3. Exploratory Data Analysis

The exploratory phase revealed that readmission risk is not randomly distributed but concentrated within specific patient profiles. This became immediately evident when examining the distribution of the target variable.

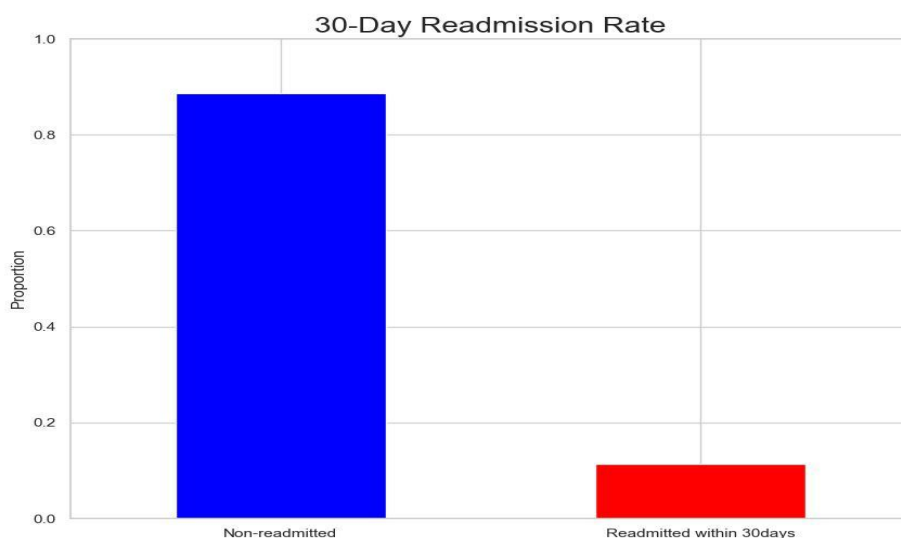


Figure 1: Class Imbalance

Only about **11–12% of patients were readmitted within 30 days**, confirming a highly imbalanced classification problem. This observation directly shaped the modelling strategy, reinforcing the need to prioritize recall over accuracy.

Further exploration of healthcare utilization variables revealed a distinct pattern. Most patients exhibited minimal interaction with the healthcare system, while a small subset showed significantly higher utilization.

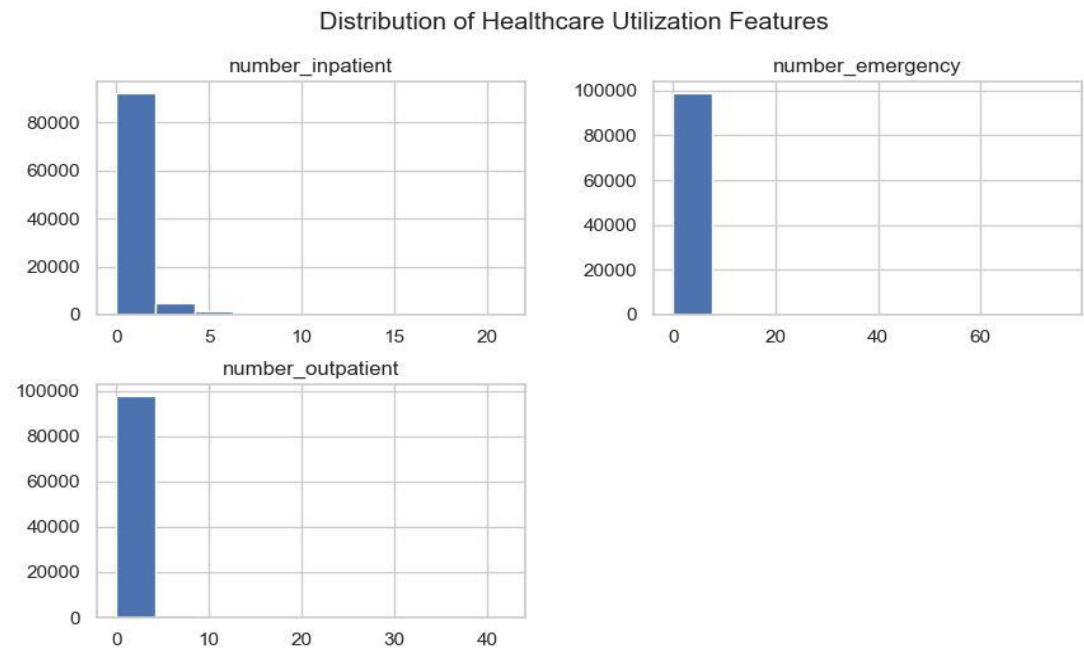


Figure 2: Healthcare Utilization Distribution

This skewed distribution indicates that risk is concentrated among a relatively small group of patients. When these variables were analysed in relation to readmission outcomes, a clear separation emerged.

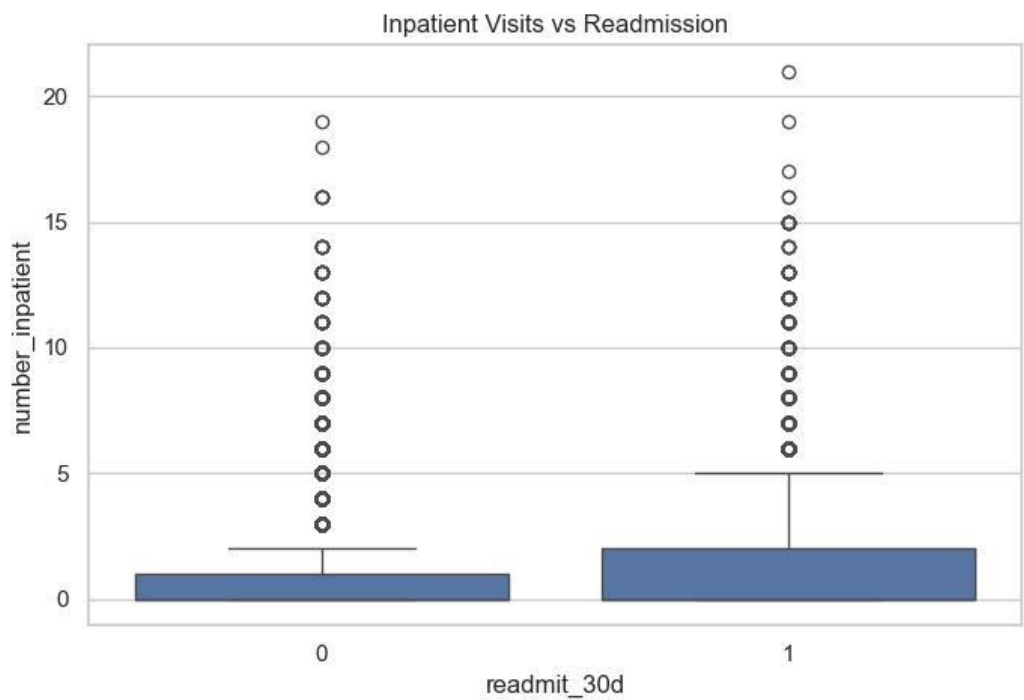


Figure 3: Utilization vs Readmission

Patients with higher inpatient visits were significantly more likely to be readmitted, establishing healthcare utilization as a primary driver of risk.

A broader structural view of the dataset further revealed strong relationships among utilization-related variables and redundancy among others.

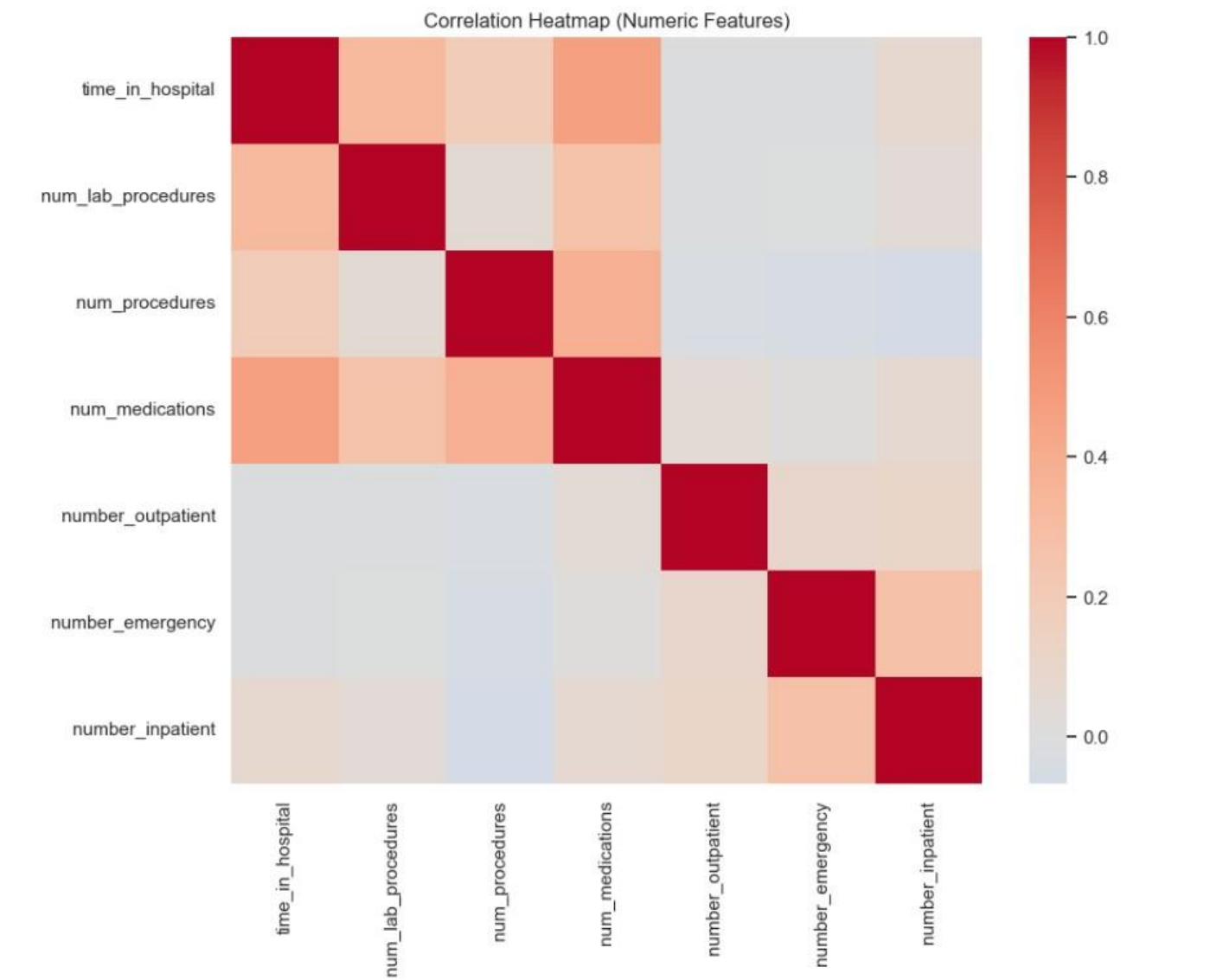


Figure 4: Correlation Heatmap

This reinforced an important modelling insight: not all variables contribute equally, and meaningful prediction requires focusing on strong signals rather than feature volume.

4. Preprocessing and Feature Selection

The preprocessing stage translated insights from EDA into a structured modelling dataset. The feature space was reduced from **114 to 84 relevant variables**, removing redundancy while preserving predictive strength.

Categorical variables were encoded using one-hot encoding, and numerical features were standardized to ensure consistent scaling. The dataset was split into training and test sets prior to transformation, ensuring unbiased evaluation.

Given the earlier identification of class imbalance, both SMOTE and class weighting were applied. This combination reinforced the model’s ability to detect minority class instances and aligned directly with the recall-focused objective.

5. Advanced Feature Engineering

Beyond preprocessing, additional features were introduced to capture interactions between variables. These included total healthcare visits, high-utilizer indicators, and composite severity measures combining hospitalization duration and diagnostic complexity.

Weighted utilization features emphasized the importance of inpatient visits, while medication pressure captured treatment intensity relative to hospital stay duration. These transformations reflect a deeper understanding that patient risk is not driven by isolated variables but by interactions between them.

6. Modelling Approach

Four models were developed to capture different learning behaviours: Logistic Regression, Random Forest, Gradient Boosting, and XGBoost. Each model contributed to understanding how different algorithms interpret the structure of the data.

Rather than relying solely on model complexity, the selection process focused on how well each model aligned with the recall-driven objective and the interaction-heavy feature space.

7. Model Evaluation: Performance in Context

Model performance was evaluated using recall, precision, F1 score, and ROC-AUC, with recall explicitly prioritized.

Cross-validation results showed similar performance across models:

- Logistic Regression: **0.644**
- Random Forest: **0.613**
- Gradient Boosting: **0.646**
- XGBoost: **0.642**

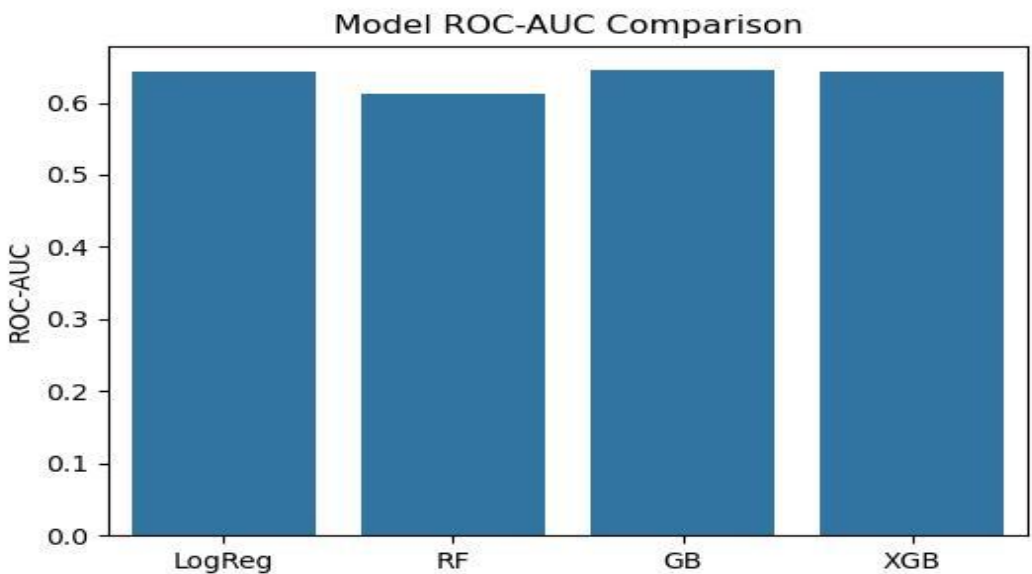


Figure 5: Model ROC -AUC Comparison

Although Gradient Boosting achieved the highest ROC-AUC, the differences were minimal. XGBoost was selected due to its robustness, ability to capture nonlinear relationships, and compatibility with engineered features.

8. Threshold Optimization: Aligning with Real-World Objectives

Default classification thresholds were insufficient for this problem. Using predicted probabilities, thresholds were systematically evaluated to understand the trade-off between recall and precision.

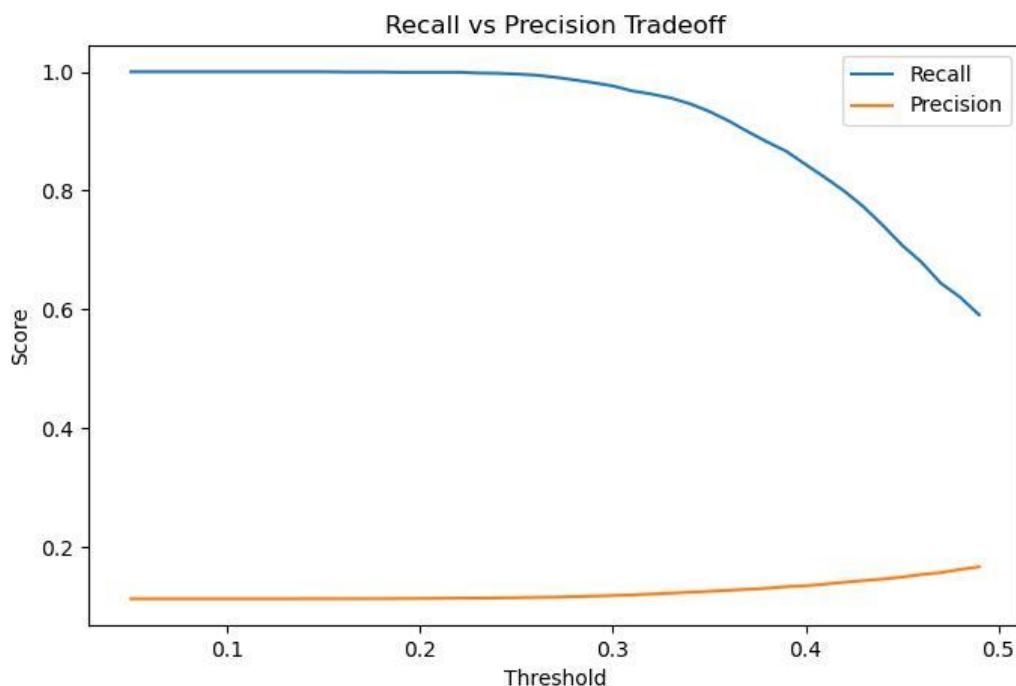


Figure 5: Recall vs Precision Trade-off

At low thresholds, recall reached **100%**, but precision remained low (~11%). As thresholds increased, recall decreased while precision improved.

At the selected threshold of **0.45**, the model achieved:

- Recall: **0.71**
- Precision: **0.15**
- F1 Score: **0.25**
- ROC-AUC: **0.649**

This threshold provided the best balance between identifying high-risk patients and maintaining manageable false positives.

9. Final Model Evaluation and Interpretation

The final model performance demonstrates how the system behaves in practice.

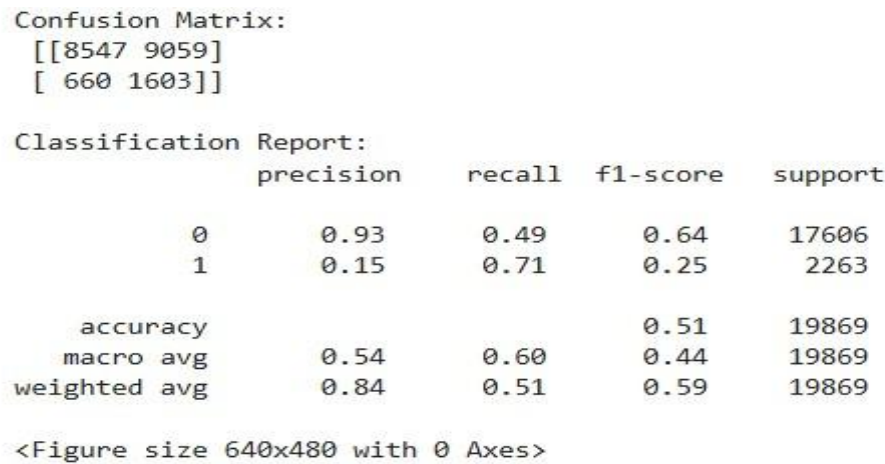


Figure 6: Confusion Matrix and Classification Report

The model successfully identifies 71% of high-risk patients, significantly reducing missed cases. While precision remains low, this is expected given the imbalance and aligns with the project objective.

10. Feature Importance and Validation of Insights

Feature importance analysis confirms that the model’s behaviour aligns with earlier findings.

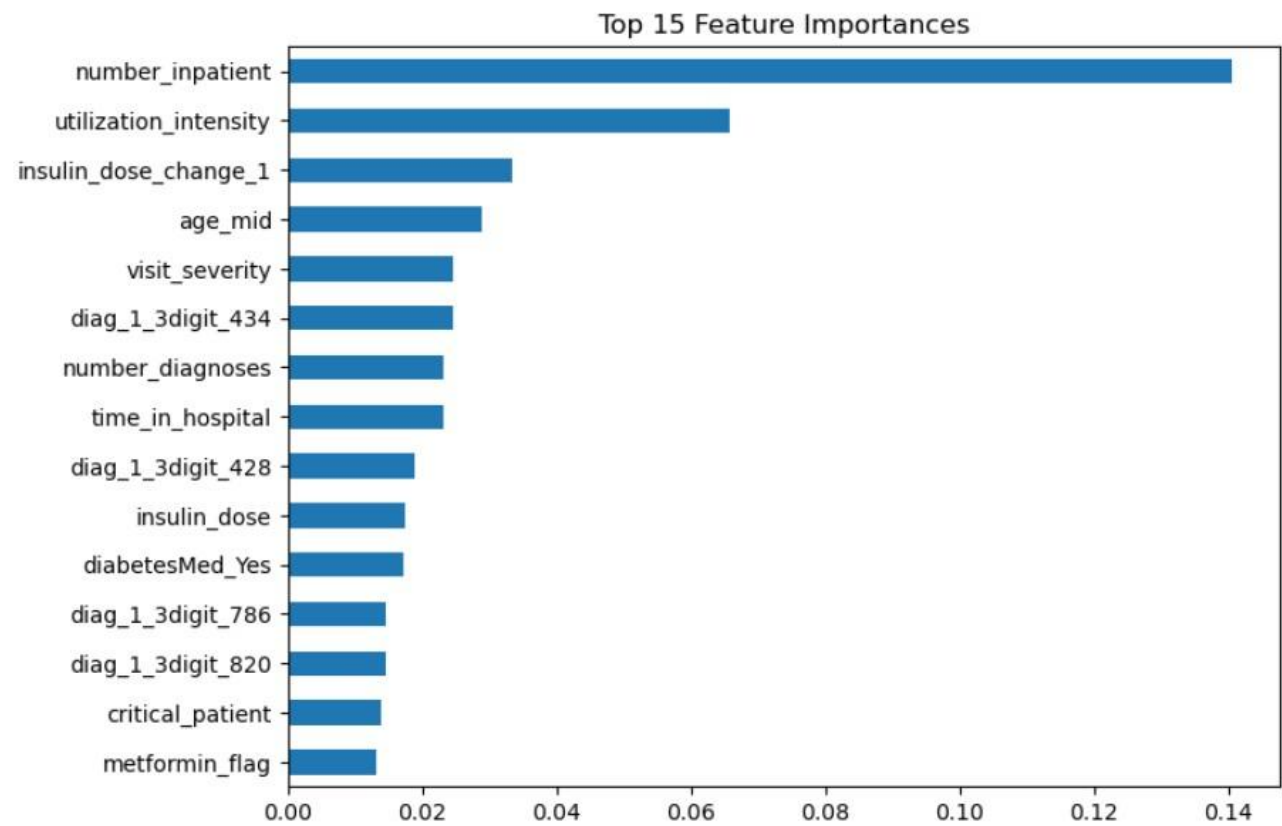


Figure 7: Top 15 feature importance

The most influential features include healthcare utilization, severity indicators, and medication-related variables. Notably, engineered interaction features rank highly, validating the decision to model relationships between variables.

11. Does the Model Solve the Problem?

The objective of this project was to reduce missed high-risk patients, not to maximize overall accuracy. By identifying approximately **71% of patients who will be readmitted**, the model significantly improves early risk detection.

While this comes at the cost of increased false positives, the trade-off is appropriate in a healthcare context where early intervention can prevent costly readmissions.

12. Limitations

Despite strong performance, limitations remain. The dataset is highly imbalanced, which inherently constrains precision. Some clinical variables were excluded due to missingness, and the absence of time-series data limits the ability to capture patient progression.

Additionally, the model has not been validated on external datasets, which may affect generalizability.

13. Future Work

Future improvements could include incorporating time-series patient data, validating the model across multiple hospital systems, and integrating SHAP-based explainability into clinical workflows. Cost-sensitive learning could further refine the balance between recall and precision.

14. Conclusion

This project demonstrates that hospital readmissions among diabetic patients are predictable when data is transformed into meaningful clinical signals and analyzed using a recall-driven approach.

By combining rigorous preprocessing, advanced feature engineering, and a carefully tuned XGBoost model, the analysis provides a practical decision-support tool capable of improving patient outcomes and reducing healthcare costs.